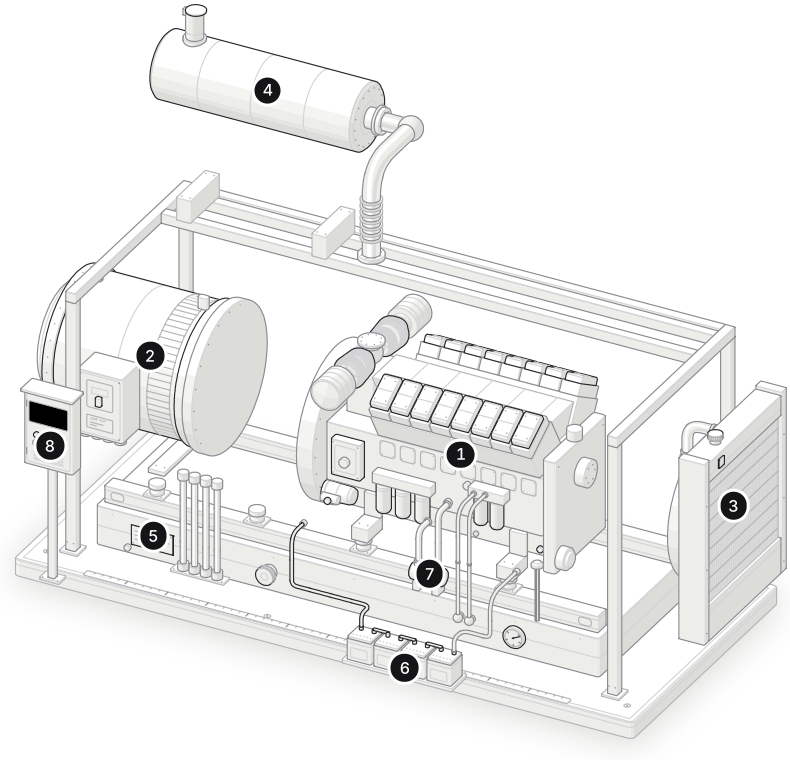
EXPLAINED DRAWING

Standby generator: how a data centre keeps the lights on

DRAWING
AN-002



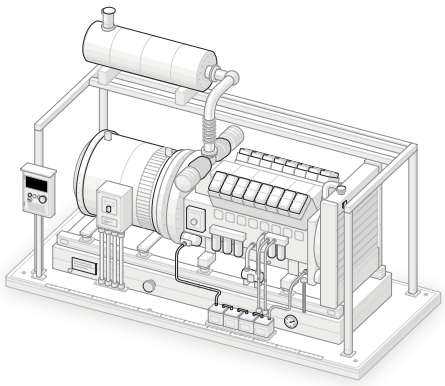
PARTS

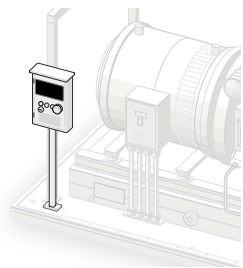
01 Diesel engine
02 Alternator
03 Radiator and fan
04 Exhaust silencer
05 Base fuel tank and mounts
06 Starter batteries
07 Jacket-water heater
08 Control panel

SPECIFICATION

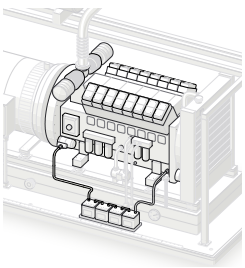
Rating	2,500 kVA / 2,000 kW
Output	400 V, 50 Hz, 3,608 A a phase
Engine	V16 diesel, 1,500 rpm
Start	24 V electric, about 150 rpm
Transfer	8.8 s (limit 10 s)
Fuel	5,500 L, about 424 L/h at 1,600 kW
Standby	coolant held at 40 °C

ASSEMBLED

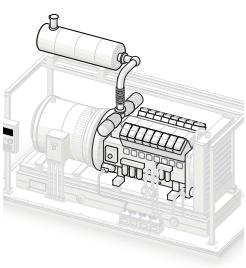




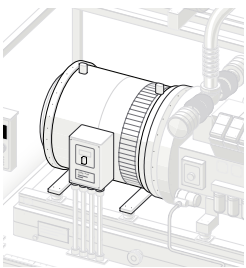
01
Mains fails



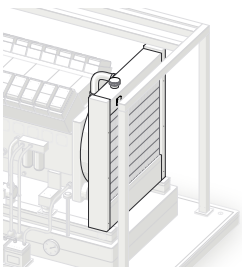
02
Batteries crank



03
Engine fires



04
On load



05
Cooling the engine